

MODULAR Reference Guide

MAPI-02

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Preface

About This Guide

This guide describes the commands used within the Modular API – a unified facade server that provides centralized management for multiple modules through a single entry point. It enables unified authentication, authorization, and audit capabilities across all integrated services.

Target Audience

This guide is designed for:

- **System Administrators** responsible for deploying and managing Modular API infrastructure
- **Security Engineers** configuring policies, groups, and user access controls
- **DevOps Engineers** integrating Modular API into CI/CD pipelines
- **Developers** extending Modular API with custom modules

Related Documents

- For Modular API setup and usage, see Modular API User Guide
- For Modular CLI configuration and usage, see [Modular-CLI README](#)
- For Modular SDK integration, see [Modular-SDK Documentation](#)
- For M3 Admin CLI usage and commands, refer to M3 Admin User Guide

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 - [Command] install - Install module
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 - [Command] run - Starts modular server
 - [Command] uninstall - Uninstall module
 - [Command] validate - Validate module CLI structure without installing. Checks naming conventions, hierarchy integrity, group/command consistency, and modular-api commands_generator compatibility. Use before 'modular in...

Detailed Command Reference

Group: modular

Description:

Configuration settings for modular-api

Usage:

```
modular [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
-v, --version	flag	X	False	Show the version and exit.

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Group: modular group

Description:

Manages groups which using to define user permissions

Usage:

```
modular group
```

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Command: modular group add

Description:

Adds group with allowed actions based on provided policy(ies) name(s)

Usage:

```
modular group add [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--group, -g	string	✓	-	Group name
--policy, -p	string	✓	-	Policy name that will be attached to the group
--table	flag	X	False	Use this parameter to show command's response in a Table view
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Command: modular group add_policy

Description:

Adds policies to already existing group

Usage:

```
modular group add_policy [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--group, -g	string	✓	-	Group name
--policy, -p	string	✓	-	Policy name that will be attached to the group
--table	flag	✗	False	Use this parameter to show command's response in a Table view
--json	flag	✗	False	Use this parameter to show command's response in a JSON view

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Command: modular group delete**Description:**

Deletes group

Usage:

```
modular group delete [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--group, -g	string	✓	-	Group name
--table	flag	✗	False	Use this parameter to show command's response in a Table view
--json	flag	✗	False	Use this parameter to show command's response in a JSON view

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Command: modular group delete_policy**Description:**

Deletes policies from already existing group

Usage:

```
modular group delete_policy [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--group, -g	string	✓	-	Group name
--policy, -p	string	✓	-	Policy name that will be detached from the group
--table	flag	✗	False	Use this parameter to show command's response in a Table view
--json	flag	✗	False	Use this parameter to show command's response in a

Option	Type	Req	Default	Description
				JSON view

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Command: *modular group describe*

Description:

Describes specified group or list all groups

Usage:

```
modular group describe [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--group, -g	string	X	-	Group name. If not specified then all existing groups willbe listed
--table	flag	X	False	Use this parameter to show command's response in a Table view
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Group: modular policy

Description:

Manages policies documents which define groups permissions

Usage:

```
modular policy
```

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Command: *modular policy add*

Description:

Creates a policy that defines the permissions you can assign to a group.

Usage:

```
modular policy add [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--policy, -p	string	✓	-	Policy name
--policy_path, -path	string	✓	-	Path to policy document. File extension: *.json
--table	flag	✗	False	Use this parameter to show command's response in a Table view
--json	flag	✗	False	Use this parameter to show command's response in a JSON view

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Command: *modular policy delete*

Description:

Deletes policy by provided name

Usage:

```
modular policy delete [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--policy, -p	string	✓	-	Permission policy name
--table	flag	✗	False	Use this parameter to show command's response in a Table view
--json	flag	✗	False	Use this parameter to show command's response in a JSON view

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Command: *modular policy describe*

Description:

Describes permissions defined in policy.

Usage:

```
modular policy describe [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--policy, -p	string	X	-	Policy name. All existed policies will be described in case if parameter not provided
--expand, -E	flag	X	False	Specify to describe policies with content. Has no effect and always 'True' if '--policy' parameter passed
--table	flag	X	False	Use this parameter to show command's response in a Table view
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Command: modular policy update

Description:

Rewrites policy permissions.

Usage:

```
modular policy update [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--policy, -p	string	✓	-	Policy name
--policy_path, -path	string	X	-	Path to policy document. File extension: *.json
--table	flag	X	False	Use this parameter to show command's response in a Table view
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Group: modular user

Description:

Manages users that can submit request to server

Usage:

```
modular user
```

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Command: modular user add

Description:

Add user to white list

Usage:

```
modular user add [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--username, -u	string	✓	-	Username that will be added to trusted users
--group, -g	string	✓	-	Group name(s) that will be attached to user
--password, -p	string	✗	-	Password that will be attached to user. In case password is not provided, it will be generated automatically.
--table	flag	✗	False	Use this parameter to show command's response in a Table view
--json	flag	✗	False	Use this parameter to show command's response in a JSON view

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Command: modular user add_to_group

Description:

Adds group(s) to the user

Usage:

```
modular user add_to_group [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--username, -u	string	✓	-	Username
--group, -g	string	✗	-	Group name(s) that will be attached to user
--table	flag	✗	False	Use this parameter to show command's response in a Table view
--json	flag	✗	False	Use this parameter to show command's response in a JSON view

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Command: modular user block**Description:**

Sets user's state to "blocked"

Usage:

```
modular user block [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--username, -u	string	✓	-	Username
--reason, -rsn	string	✓	-	The textual reason of user blocking
--table	flag	✗	False	Use this parameter to show command's response in a Table view
--json	flag	✗	False	Use this parameter to show command's response in a JSON view

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Command: modular user change_password**Description:**

Changes user's password

Usage:

```
modular user change_password [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--username, -u	string	✓	-	Username
--password, -pwd	string	✓	-	New user password
--table	flag	✗	False	Use this parameter to show command's response in a Table view
--json	flag	✗	False	Use this parameter to show command's response in a JSON view

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Command: modular user change_username**Description:**

Changes user's name

Usage:

```
modular user change_username [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
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Option	Type	Req	Default	Description
--old_username, -ou	string	✓	-	Current Username
--new_username, -nu	string	✓	-	Username to set
--table	flag	X	False	Use this parameter to show command's response in a Table view
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Command: modular user delete

Description:

Deletes user from white list

Usage:

```
modular user delete [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--username, -u	string	✓	-	Username that will be deleted from white list
--table	flag	X	False	Use this parameter to show command's response in a Table view
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Command: modular user delete_meta_attribute

Description:

Delete parameter/s name from the user meta information

Usage:

```
modular user delete_meta_attribute [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--username, -u	string	✓	-	User name
--meta_type, -mt	choice	X	allowed_values	Type of meta. Supports two types: "allowed_values": restricts the parameters allowed for the user (Default). "aux_data": stores additional private user data for customized usage.. Available choices: allowed_values, aux_data
--key, -k	string	✓	-	Parameter name. Multiple values allowed (e.g. --key value1 --key value2).
--table	flag	X	False	Use this parameter to show command's response in a Table view

Option	Type	Req	Default	Description
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Command: modular user describe

Description:

Describes user(s) information

Usage:

```
modular user describe [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--username, -u	string	X	-	Username. If not specified - all existing users will be listed
--table	flag	X	False	Use this parameter to show command's response in a Table view
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Command: modular user get_meta

Description:

Describe meta information for the specified user

Usage:

```
modular user get_meta [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--username, -u	string	✓	-	User name
--table	flag	X	False	Use this parameter to show command's response in a Table view
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Command: modular user remove_from_group

Description:

Removes group(s) from the user

Usage:

```
modular user remove_from_group [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--username, -u	string	✓	-	Username
--group, -g	string	X	-	Group name(s) that will be detached from user
--table	flag	X	False	Use this parameter to show command's response in a Table view
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Command: modular user reset_meta**Description:**

Erase all meta information from the user item

Usage:

```
modular user reset_meta [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--username, -u	string	✓	-	User name
--table	flag	X	False	Use this parameter to show command's response in a Table view
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Command: modular user set_meta_attribute**Description:**

Add or replace Modular user meta information.

Usage:

```
modular user set_meta_attribute [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--username, -u	string	✓	-	Username
--meta_type, -mt	choice	X	allowed_values	Type of meta. Supports two types: "allowed_values": restricts the parameters allowed for the user (Default). "aux_data": stores additional private user data for customized usage.. Available choices: allowed_values, aux_data
--key, -k	string	✓	-	Parameter name
--value, -v	string	X	-	Parameter value. Multiple values allowed (e.g. -value value1 -value value2).

Option	Type	Req	Default	Description
--value_as_json, -vj	string	X	-	Parameter value as json (e.g. {"k": "v"} or ["v1", "v2"]).
--table	flag	X	False	Use this parameter to show command's response in a Table view
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Command: modular user unblock

Description:

Resets user's state from blocked

Usage:

```
modular user unblock [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--username, -u	string	✓	-	Username
--reason, -rsn	string	✓	-	The textual reason of user unblocking
--table	flag	X	False	Use this parameter to show command's response in a Table view
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Command: modular user update_meta_attribute

Description:

Update Modular user meta information.

Usage:

```
modular user update_meta_attribute [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--username, -u	string	✓	-	User name
--meta_type, -mt	choice	X	allowed_values	Type of meta. Supports two types: "allowed_values": restricts the parameters allowed for the user (Default). "aux_data": stores additional private user data for customized usage.. Available choices: allowed_values, aux_data
--key, -k	string	✓	-	Parameter name
--value, -v	string	X	-	Parameter value. Multiple values allowed (e.g. --value value1 --value value2).
--value_as_json, -vj	string	X	-	Parameter value as json (e.g. {"k": "v"} or ["v1", "v2"]).

Option	Type	Req	Default	Description
--table	flag	X	False	Use this parameter to show command's response in a Table view
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Command: modular audit**Description:**

Describes audit

Usage:

```
modular audit [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--group, -g	string	X	-	Filter by group name. If “--group” and “--from_date” and “--to_date” parameters not specified - will be shown all events for the last 7 days
--command, -c	string	X	-	Filter by command name
--from_date, -fd	string	X	-	Filter by date from which records are displayed. Format yyyy-mm-dd
--to_date, -td	string	X	-	Filter by date until which records are displayed. Format yyyy-mm-dd
--limit, -l	integer (1 to 100)	X	10	Number of records that will be shown. Default value is 10. Will have no effect if “--group” not specified
--invalid, -I	flag	X	False	Flag to show only invalid audit events.
--table	flag	X	False	Use this parameter to show command’s response in a Table view
--json	flag	X	False	Use this parameter to show command’s response in a JSON view

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Command: modular describe

Description:

Describe all available modules versions and Modular-SDK/CLI version.

Usage:

```
modular describe [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--table	flag	X	False	Use this parameter to show command's response in a Table view
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Command: modular get_stats

Description:

Saves usage statistic to a file

Usage:

```
modular get_stats [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--from_month, -fm	string	X	-	Filter by month from which records are displayed. Format "yyyy-mm". If not specified - current month will be used
--to_month, -tm	string	X	-	Filter by month until which records are displayed. Format yyyy-mm. If not specified - next month will be used
--display_table, -D	flag	X	False	Flag to show report in terminal. If not specified - report will be stored into the file
--path, -p	string	X	-	Directory path to saving report file. If not specified - report will be saved in user home directory
--table	flag	X	False	Use this parameter to show command's response in a Table view
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Command: modular install

Description:

Install module

Usage:

```
modular install [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--module_path, -path	string	✓	-	Path to needed tool source files
--force, -f	flag	X	False	Install module even if CLI validation errors are found
--recommendations, -rec	flag	X	False	Show additional code quality recommendations during pre-install validation
--table	flag	X	False	Use this parameter to show command's response in a Table view
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Command: modular policy_simulator

Description:

Policy simulator

Usage:

1. Get action status by user and command `modular policy_simulator --user $user_name --command "admin aws add_image"`
2. Get action status by group and command `modular policy_simulator --group $group_name --command "admin aws add_image"`
3. Get action status by policy and command `modular policy_simulator --policy $policy_name --command "admin aws add_image"`

Usage:

`modular policy_simulator [OPTIONS]`

Options:

Option	Type	Req	Default	Description
<code>--user, -u</code>	string	X	-	User which permissions will be checked
<code>--group, -g</code>	string	X	-	Group which permissions will be checked
<code>--policy, -p</code>	string	X	-	Policy which permissions will be checked
<code>--command, -cmd</code>	string	✓	-	* Command call string to be checked
<code>--table</code>	flag	X	False	Use this parameter to show command's response in a Table view
<code>--json</code>	flag	X	False	Use this parameter to show command's response in a JSON view

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Command: modular run

Description:

Starts modular server

Usage:

```
modular run [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--host, -h	string	X	0.0.0.0	Host to start the server
--port, -p	integer	X	8085	Host to start the server
--prefix, -pr	string	X	"	Global api prefix. By default there is no prefix
--gunicorn, -g	flag	X	False	Whether to run the server using gunicorn
--workers, -nw	integer	X	33	Number of gunicorn workers. Has effect only if --gunicorn flag is set
--worker_timeout, -wt	integer	X	0	Gunicorn worker timeout in seconds. By default there is no timeout
--swagger	flag	X	False	Whether to server swagger
--swagger_prefix	string	X	/swagger	Swagger path prefix

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Command: modular uninstall

Description:

Uninstall module

Usage:

```
modular uninstall [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
--module_name, -name	string	✓	-	Name of the module to be uninstalled
--table	flag	X	False	Use this parameter to show command's response in a Table view
--json	flag	X	False	Use this parameter to show command's response in a JSON view

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Command: modular validate

Description:

Validate module CLI structure without installing.

Checks naming conventions, hierarchy integrity, group/command consistency, and modular-api commands_generator compatibility.

Use before 'modular install' to catch structural issues early, or in CI/CD pipelines with `--strict` to gate merges.

Examples:

1. Basic validation: `modular validate --module_path /path/to/module`
2. Strict mode for CI (warnings = failure): `modular validate --module_path /path/to/module --strict`
3. With code quality recommendations: `modular validate --module_path /path/to/module --recommendations`

Usage:

```
modular validate [OPTIONS]
```

Options:

Option	Type	Req	Default	Description
<code>--module_path, -path</code>	string	✓	-	Path to the module to validate (same path as for install)
<code>--strict, -s</code>	flag	X	False	Treat warnings as errors (exit code 1)
<code>--recommendations, -rec</code>	flag	X	False	Include additional code quality recommendations (type hints, return types, param alignment, etc.)
<code>--table</code>	flag	X	False	Use this parameter to show command's response in a Table view
<code>--json</code>	flag	X	False	Use this parameter to show command's response in a JSON view

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Version History

Version	Date	Summary

View Changelog: <https://git.epam.com/epmc-eoos/m3-modular-admin/-/blob/develop/CHANGELOG.md>

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